Language-Gated GR-ConvNet: Steering a Compact Grasp CNN with Frozen CLIP Patch–Text Similarity

Minh-Ngoc Luong
Hanoi University of Science and Technology
ngoclm004@gmail.com

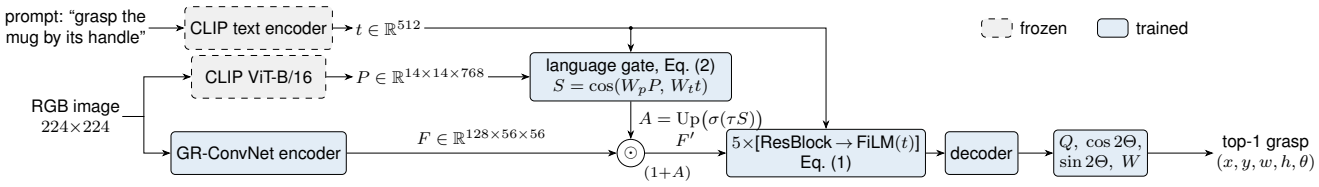


Figure 1. **Language-Gated GR-ConvNet.** Blue: trained (2.9M parameters); CLIP [5]: frozen. The prompt enters through a spatial gate A and through FiLM [4] after every residual block.

Abstract

Language-driven grasp detection places a grasp on the object part named by a prompt. The proposed method steers the GR-ConvNet grasp CNN with FiLM and a language gate*: a spatial map of similarities between projections of frozen CLIP patch tokens and of the prompt embedding. On the annotated pairs of the official LGD test split of Grasp-Anything++, the model reaches 68.6% seen and 57.7% unseen success (two-seed mean), against 54.8% and 40.7% with FiLM alone. A control gate with the same CLIP features but no prompt scores 11.6 and 11.8 unseen points lower in the two seeds ($p < 10^{-7}$); prompt-dependent gating gives 9.4 of the 16.0-point gain in harmonic mean over FiLM. Code:* https://github.com/Ngoc-LM/language-gated-grasp.

1. Introduction

A parallel-jaw grasp [1] is a rotated rectangle $g = (x, y, w, h, \theta)$. Language-driven grasp detection conditions g on a prompt such as "grasp the knife by its handle", so the model must also locate the named part. On Grasp-Anything++ (1M images, 10M prompts), the reference method LGD [8] is a diffusion model. This work tests whether a single-pass grasp CNN can instead be steered by language. The contributions are (i) a *language gate* that re-weights GR-ConvNet [2] features by CLIP [5] patch–text similarity (Fig. 1); (ii) an ablation (text-free gate, two seeds, paired tests) that separates CLIP features from prompt-dependent gating.

2. Related Work

Generative grasp CNNs [2, 3] regress dense grasp maps. CROG [6] uses CLIP for referring grasp synthesis, LGD [8] denoises grasps under text guidance on data built with foundation models [9], and mask-guided attention [7] adds spatial language grounding to it. Pixel–text CLIP maps drive dense prediction [10]; the gate uses one as a spatial prior.

3. Method

Grasp maps. As in [2], the network predicts four 224×224 maps: quality Q, $\cos 2\Theta$, $\sin 2\Theta$ and opening W. Targets fill the central third of each ground-truth rectangle. The grasp is read at the most interior pixel of the near-maximum plateau of the smoothed Q, with $\theta = \frac{1}{2}\,\mathrm{atan2}(\sin 2\Theta, \cos 2\Theta)$ and $h = w/2$.

FiLM. The encoder maps the image to $F \in \mathbb{R}^{128\times56\times56}$, followed by five residual blocks and a decoder. With t the normalised CLIP text embedding, each residual block k is followed by

$$F \leftarrow \big(1 + \Delta\gamma_k(t)\big) \odot F + \beta_k(t), \tag{1}$$

where $\Delta\gamma_k, \beta_k \in \mathbb{R}^{128}$ come from a zero-initialised linear layer, so each FiLM layer is the identity at initialisation.

Language gate (proposed). FiLM applies the same channel modulation at every pixel, so it can favour one object among several only indirectly. The language gate therefore compares the 14×14 post-LayerNorm patch tokens P of a frozen CLIP ViT-B/16 with the prompt and re-weights the encoder output:

$$\begin{aligned} S_{ij} &= \cos(W_p\, p_{ij},\, W_t\, t), \\ A = \mathrm{Up}_{56}\big(\sigma(\tau S)\big), &\qquad F' = F \odot (1 + A), \end{aligned} \tag{2}$$

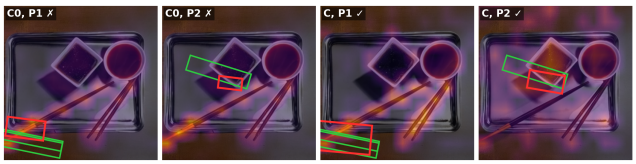


Figure 2. **Gate maps** (heat), top-1 (red), GT (green) for P1 "Hold chopsticks at its pair of sticks" and P2 "Take hold of bottle on its container" (unseen; largest C-gate change where only C solves both): the C0 gate ignores the prompt, the C gate follows it.

	Model	Seen	Unseen	H	ms
A	GR-ConvNet (RGB only)	27.3	27.8	27.5	4.7
B	+ FiLM(t)	$54.8_{\pm 2.7}$	$40.7_{\pm 0.6}$	$46.7_{\pm 1.4}$	14.1
C0	B + text-free gate	$63.3_{\pm 0.5}$	$46.0_{\pm 1.9}$	$53.3_{\pm 1.1}$	19.6
C	B + language gate	$68.6_{\pm 0.4}$	$57.7_{\pm 1.8}$	$62.7_{\pm 1.3}$	19.7
D	C + gate supervision	70.8	57.7	63.6	21.5

Table 1. **Test success (%)** and T4 latency (ms, incl. CLIP). B, C0, C: two-seed mean±half-range; shaded: proposed.

where $p_{ij} \in \mathbb{R}^{768}$ is a patch token, W_p and W_t are learned affine maps to $\mathbb{R}^{256}$, τ is a learned scale (initially 10), and Up_{56} bilinearly upsamples the 14×14 map to 56×56. As $A \in (0, 1)$, $1+A$ never suppresses features. The **C0 control** replaces $W_t t$ with a learned vector q: its gate sees the same CLIP features but not the prompt (FiLM still receives t), and it has 131k fewer parameters than C. **Loss:** Smooth-ℓ_1 summed over the four maps. Variant D adds $0.5\,\mathrm{BCE}(\tau S, M)$, M the pooled ground-truth Q mask.

4. Experiments

Protocol. Training uses 35,927 Grasp-Anything++ scenes (183,559 prompt–grasp pairs) and validation 734 scenes. Testing follows the official LGD seen/unseen split, restricted to annotated pairs: 1,376 seen and 601 unseen from 727 of 803 object keys. Test scenes, and training pairs whose object name is an unseen test name, are excluded from training; a word search finds unseen names in 5,332 training pairs, all as parts of other objects, and 40 of the 41 seen names occur in training. A top-1 grasp is correct if its IoU exceeds 0.25 and its angle error is below 30° for some ground-truth grasp of the prompted part; H is the harmonic mean of seen and unseen success. All runs use AdamW (lr 10^{-3}, weight decay 10^{-4}, 500 warm-up steps, cosine), batch 32 and 6 epochs on one T4; checkpoints are selected on 1,000 validation pairs.

Results. FiLM (B) raises H by 19.2 points, CLIP features behind the text-free gate (C0) add 6.6, and letting the gate read the prompt (C) adds 9.4. Seed for seed, C exceeds C0 by 11.6/11.8 unseen and 6.3/4.4 seen points; paired McNemar tests give $p < 10^{-3}$ in all four comparisons, and scene-resampled 95% bootstrap intervals exclude zero. The unseen gain of C0 over B varies between seeds ($+7.8$, $p < 10^{-3}$; $+2.8$, $p = 0.21$), whereas the C0→C gain is stable. In the 118 multi-object scenes, C solves two prompts on different objects in 55, C0 in 40 (seed 0, Fig. 2). Prompt-dependent gating thus gives 69% of the B→C gain on unseen objects, 59% in H and 38% on seen ones. D does not differ significantly from C ($p \geq 0.06$).

Cost and limitations. Over B, C adds 0.33M trained parameters and the frozen 86M CLIP image encoder (13.0 ms/frame with a cached prompt). A already exceeds the 18% unseen reported for GR-ConvNet with CLIP [8], so LGD's 48/42% are not compared.

5. Conclusion

The language gate raises the H of a FiLM-conditioned GR-ConvNet from 46.7 to 62.7. The text-free control attributes 9.4 of these 16.0 points to reading the prompt, including 69% of the unseen gain. All models train on 3.6% of the images, so the evidence is limited to this subset, two seeds and a 14×14 gate; full-data training, part-level gate supervision and the LGD protocol are the next steps.

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